Parse Deep Benchmark v1 — 2-page review brief

Date: 2026-06-11 · Status: BUILD-v0 (spec only · zero cells run · $0) · Bar: ParseBench / NeurIPS-paper, strictly above SAVIOR-Bench Approval: PENDING build-v0 review (chief-research-scientist · chief-architect · docs-techwriter · benchmark-format-steward) Companion: full spec at `PARSE-DEEP-BENCHMARK-FRAMEWORK-2026-06-11-v1.md` · dashboard at `parse-deep-build-v0/index.html`

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§1 — What this is, in one paragraph

A pre-registered, third-party-anchored, ensemble-aware framework to defensibly answer **"is the HyperAPI parse primitive the best general parsing primitive across finance documents."** 50 dimensions across image / text / doc-type / language / density / CV / enterprise / NLP / print / repro. A 60-row commercial-vs-OSS model panel. A 16-type × 50-dim matrix (800 cell-classes). Engineered to **succeed whether HyperAPI parse wins or loses on any dimension** — researcher-not-advocate. Inherits SAVIOR-Bench Appendix D 4-part cell contract; lifts the bar to ParseBench-style per-rule = per-doc score vectors, paired-bootstrap CIs, Holm-Bonferroni multiplicity correction, contamination + held-out audit, NeurIPS reproducibility checklist.

§2 — All 51 metrics (A–K · 50 v1 + D-51 UEM v1.1 amendment 2026-06-15)

Dim	Cat	Name	Metric
D-01	A-IMAGE	IMG-DPI	`effective_dpi + cer_delta_strat`
D-02	A-IMAGE	IMG-MTF	`mtf_50_lp_mm + cer_delta`
D-03	A-IMAGE	IMG-ILLUMINATION	`illumination_uniformity_pct + cer_delta + bbox_iou_delta`
D-04	A-IMAGE	IMG-DEWARP	`dewarp_rmse_px + cer_delta`
D-05	A-IMAGE	IMG-SKEW	`skew_recovery_pct_cer`
D-06	A-IMAGE	IMG-JPEG-Q	`cer_delta_q40_vs_q90`
D-07	B-TEXT	OCR-RECALL	`savior_word_recall`
D-08	B-TEXT	OCR-PRECISION	`savior_word_precision + hallucinated_token_count`
D-09	B-TEXT	OCR-EDIT-DISTANCE	`ocr_edit_distance_norm`
D-10	B-TEXT	OCR-WER	`wer`
D-11	B-TEXT	OCR-CER	`cer_per_script`
D-12	B-TEXT	OCR-DIGIT-MICR-FIDELITY	`digit_micr_fidelity_f1`

Dim	Cat	Name	Metric
D-13	C-DOCTYPE	DT-AP-AR	`f1_per_type`
D-14	C-DOCTYPE	DT-PAYMENTS	`f1_per_type + per_field`
D-15	C-DOCTYPE	DT-CAPITAL-MARKETS	`f1_per_type`
D-16	C-DOCTYPE	DT-ADJACENT	`f1_per_type`
D-17	C-DOCTYPE	DT-DENSITY	`metric_per_density_stratum`
D-18	C-DOCTYPE	DT-MULTI-COL	`multi_col_layout_f1 + column_count_err`
D-19	C-DOCTYPE	DT-NESTED-TABLE	`nested_table_cell_f1`
D-20	D-LANGUAGE	LANG-CER-PER-SCRIPT	`cer_per_script`
D-21	D-LANGUAGE	LANG-KIE-XFUND	`kie_f1_per_language`
D-22	D-LANGUAGE	LANG-MIXED-SCRIPT	`cer_mixed_script`
D-23	E-DENSITY	LAYOUT-ELEM-ACCURACY	`layout_element_accuracy`
D-24	E-DENSITY	LAYOUT-READING-ORDER	`reading_order_rank_correlation`
D-25	E-DENSITY	LAYOUT-PAIRS	`pairs_layout_zeuclidean`
D-26	F-CV	BBOX-IOU-TOKEN	`bbox_iou_token + match_rate`
D-27	F-CV	BBOX-FIELD-TO-REGION	`field_to_region_grounding_acc_iou50`
D-28	F-CV	LAYOUT-COCO-MAP	`coco_mAP_50_75_95 + per_class_AP`

Dim	Cat	Name	Metric
D-29	F-CV	SEG-IOU	`seg_pixel_iou`
D-30	F-CV	VLM-PAGE-IMAGE-FIDELITY	`psnr + ssim`
D-31	G-ENTERPRISE	TABLE-GTRM	`grits + tablerecordmatch + gtrm`
D-32	G-ENTERPRISE	TABLE-TEDS	`teds`
D-33	G-ENTERPRISE	TABLE-CELL-F1	`cell_content_f1 + per_row_type`
D-34	G-ENTERPRISE	KIE-FUNSD-F1	`entity_f1`
D-35	G-ENTERPRISE	KIE-CORD-F1	`entity_f1`
D-36	G-ENTERPRISE	DOWNSTREAM-LIFT	`parse_lift_paired_bootstrap`
D-37	H-NLP	PARSE-INTENT-F1	`parse_intent_f1 + wer`
D-38	H-NLP	VQA-ANLS	`anls`
D-39	H-NLP	SEMANTIC-BERTSCORE	`bertscore_f1`
D-40	H-NLP	INTENT-CLASSIFICATION	`classify_accuracy_per_type`
D-41	I-PRINT	LAT-P50	`latency_p50_ms`
D-42	I-PRINT	LAT-P95-P99	`latency_p95_ms + latency_p99_ms`
D-43	I-PRINT	COST-DOLLAR-PER-CORRECT	`dollars_per_correct_call`
D-44	I-PRINT	FAILURE-RATE	`failure_rate_pct + per_mode_tally`
D-45	I-PRINT	PAGE-SCALING	`cer_per_page_band + timeout_rate`

Dim	Cat	Name	Metric
D-46	J-REPRO	STAT-CI95	`ci95_bootstrap_2000_seed_20260516`
D-47	J-REPRO	STAT-PAIRED-CI	`paired_bootstrap_ci`
D-48	J-REPRO	STAT-IAA	`cohens_kappa + krippendorff_alpha`
D-49	J-REPRO	REPRO-FACTORY-GREEN	`factory_reproduced_bool`
D-50	J-REPRO	REPRO-CONTAMINATION	`contamination_audit_pass_bool`
D-51	**K-UEM**	**UEM-UNIVERSAL-OCR-METRIC**	`uem_score` over {HOF · Markdown · HTML · Tokens+BB} · cross-format judge=GPT-5.4-mini (Azure) · same-format model-free

Bijection to the 2026-05-27 9-D spec preserved (every D-OCR/D-PARSE-INTENT/D-LAYOUT/D-BBOX/D-TABLE/D-DOC-QUALITY/D-LANG/D-DOWNSTREAM/D-COST-LATENCY maps to a D-XX). No dimension dropped.

§3 — The 130-row model panel

56 commercial · 66 OSS · 8 stretch OSS = 130 rows. Floor cleared by ~5×.

Bucket	Count	Examples
Frontier closed-source LLM/VLM	13	Claude Opus 4.8 / Sonnet 4.6 / Haiku 4.5 · GPT-5.5/5.4/4.1/o3 (via Azure) · Gemini 3 Pro/Flash + 2.5 Pro · xAI Grok-4 · Cohere CR+ vision · Reka Core
Doc-AI / IDP commercial	33	Azure DI · AWS Textract · Google DocAI · LlamaParse · Reducto · **Extend Parse 2.0 (C10)** · Nanonets · Mistral OCR 2 · ABBYY · Rossum · Hyperscience · Klippa · Mindee · Veryfi · Affinda · Sensible · Instabase · Indico · IBM Watson · Oracle · Adobe · Foxit · Konfuzio · HyperVerge · Eden AI · OCR.space · Scribe · Iron OCR · Aspose · A2iA
Frontier OSS VLM-OCR	21	Chandra · MonkeyOCR + Pro-3B · PaddleOCR-VL 1.5/1.6 · MinerU 2.5/2.5-Pro/Pro · DOTS-OCR · GLM-OCR · GOT-OCR 2.0 · Qwen 3-VL-235B · Marker · Surya · Docling · Nougat · FireRed · PP-StructureV3
OSS doc-AI classics	11	TrOCR · Donut · LayoutLMv3 · LiLT · Pix2Struct · Kosmos-2.5 · Florence-2 · UDOP · DocFormerV2 · ERNIE-Layout · LayoutXLM

Bucket	Count	Examples
OSS open-weight VLMs (sweep)	21	Vary-toy · MiniCPM-V/o 2.6 · Idefics3 · CogVLM2 · InternLM-XC 2.5 · DeepSeek-VL2 · Phi-4-mm · Pixtral 12B · Molmo 72B · Aria · Llama-3.2-11B-Vision · Janus-Pro · GLM-4V · InternVL3 · Cambrian-1 · VILA · Mini-Gemini · MAmmoTH-VL · OpenELM-VL · EXAONE 3.5 VL
Table / structure / open-OCR	7	TATR · SmolDocling · olmOCR-2 · RolmOCR · TextMonkey · StructEqTable · Kraken
Hyperbots-IP `[FT]`	4	Qwen 3.6-35B-A3B `[FT]` (D-36 extractor pin) · Qwen 2.5-VL-7B/3B-F `[FT]` · hyperapi-parse-intent `[FT]`
Stretch	8	mPLUG-DocOwl 1.5 · DeepSeek-OCR-2 · Nanonets-OCR-s · LLaVA-1.6-34B · InternVL2 · Llama-3.2-90B-V · Llama-4-Maverick · Youtu-Parsing

Audit gap closed: Extend Parse 2.0 is **C10** (PENDING-KEY, gap-report contrast per Extend RealDocBench). Adapter wires fall into 3 contract shapes (class-based image-input vLLM · hosted REST multipart · OpenAI-shape chat); 7 REGISTRY entries today vs ~123 to wire at BUILD-v1.

§4 — Methodology highlights (NeurIPS-bar)

- **Pre-registration**: this framework IS the pre-registration. Every cell ships with hypothesis · acceptance · rerunnable command · `nrun_status` (default `registered-not-yet-run`) before any number lands.
- **Decoding pins** — 3 named families, machine-readable per cell:
- `parse-deep` (T=0.1, top_p=0.9, max_tokens=2048) — 48 dimensions
- `TR-000` (T=0, top_p=1, 4096) — D-31 GTRM, flagged `apples-to-oranges` per I-1
- `extract` (T=0, top_p=1, 4096) — D-36 lift, flagged `apples-to-oranges`
- **Stats**: nonparametric bootstrap, resamples=2000, seed=20260516. Paired-bootstrap on D-36 lift. Holm-Bonferroni for cross-dimension headlines (family-wise α=0.05; per-dim α≈0.001).
- **Contamination**: every `[FT]=y` row is audited vs SAVIOR-Train; unresolved overlap → cell stays `factory_reproduced=false` and renders red. Held-out 10% split per corpus.
- **Provenance**: SAVIOR-Bench Appendix D 4-part contract — `provenance · pins · rerun_command · factory_reproduced` — on every cell. `factory_reproduced=false` at adapter emit time (TR-4); only `repro_check.py` flips it true.
- **Reproducibility checklist**: 18 NeurIPS-2024 items answered item-by-item in `methodology.md §8`.
- **Comparative-study 8 invariants** (I-1 apples-to-apples ... I-8 customer-name consent): per-claim enforced; machine-greppable.

§5 — License + customer-name discipline

License matrix (framework §9.1): ParseBench Apache-2.0 (the publishability anchor) · FUNSD/CORD/PubTabNet/DocLayNet/ChartQA permissive (public) · DocVQA CC-BY-NC (internal-only) · Hyperbots-internal corpora (no customer names) · SAVIOR partial (aggregate publishable per Appendix D). Customer-name 14-roster (verbatim from `_grep_guard.sh`) build-time-grep-guarded across every artifact. PII corpora synthetic-fill or partner-consent-gated.

§6 — What's in BUILD-v0 (this package)

1. `PARSE-DEEP-BENCHMARK-FRAMEWORK-2026-06-11-v1.md` — full spec (~500 LOC, §0–§14)
2. `parse-deep-build-v0/methodology.md` — NeurIPS reproducibility checklist

3. `parse-deep-build-v0/tasks-additive-DQ-001-050.json` — 50 task entries, ADDITIVE-ONLY
4. `parse-deep-build-v0/cell-schema-v1.1.json` — JSON Schema 2020-12, backward-compat
5. `parse-deep-build-v0/index.html` — dashboard scaffold (registered-not-yet-run state)
6. `parse-deep-build-v0/BUILD-V0-LEDGER-2026-06-11.md` — acceptance-criterion mapping + grep-guard self-check

Zero cells run · $0 burned · no quota touched · no broadcasts fired.

§7 — Decisions requested from reviewer

1. **APPROVE / APPROVE-WITH-CHANGES / REJECT** on the 50-dim list — anything missing? anything to cut?
2. **Panel completeness sign-off** — 110 rows after the 2026-06-15 lit-sweep (+50 across enterprise IDP, frontier multimodal hosted, OSS doc-AI, open-weight VLMs, and table specialists). Any further additions / cuts before BUILD-v1 wires adapters?
3. **Q-DR-5** (CEO): for D-15 capital-markets — restrict HyperAPI scope to AP/AR + payments + forms, OR fund a capital-markets fine-tune to bring T-11/T-12 into the doc-type set?
4. **BLK-16 Option A (wait reset) vs Option B (paid-tier upgrade)** for HyperAPI quota — blocks the HyperAPI parse + classify columns.
5. **Greenlight BUILD-v1?** — cell execution, gated on (3) + (4) + partner-corpus consent.

Researcher-not-advocate · Hyperbots Research · 2026-06-11 · DRAFT INTERNAL until build-v0 reviewers verdict.